Aim: To Perform BI using Tableau

Theory:

Q1. Explain Tableau.

Tableau is a smart tool that turns boring numbers and data into clear, visual stories.

It connects to almost any data source you have—like Excel files, databases, or online apps—and lets you build interactive charts, maps, and dashboards just by dragging and dropping. You don't need to be a programmer.

The whole point is to help people and companies quickly see what's happening in their data, spot trends, and make smarter decisions without getting lost in complex spreadsheets.

Because it's both easy to use and powerful, Tableau helps teams ask better questions, find answers faster, and share discoveries with anyone, in a way that everyone can understand.

Q2. List Features of Tableau.

- Data Blending: Combine data from multiple sources into a single view.
- Real-Time Analysis: Connect to live data for up-to-the-second insights.
- Collaboration: Share dashboards and work together via Tableau Server/Cloud.
- Drag-and-Drop Interface: Build visualizations without coding.
- Interactive Dashboards: Create clickable, filterable reports.
- Wide Data Connectivity: Link to databases, cloud apps, spreadsheets, etc.
- Advanced Analytics: Integrate trends, forecasts, and statistical summaries.
- Mobile Responsiveness: Access and view dashboards on any device.
- Scalability: Handle large datasets and enterprise deployments.

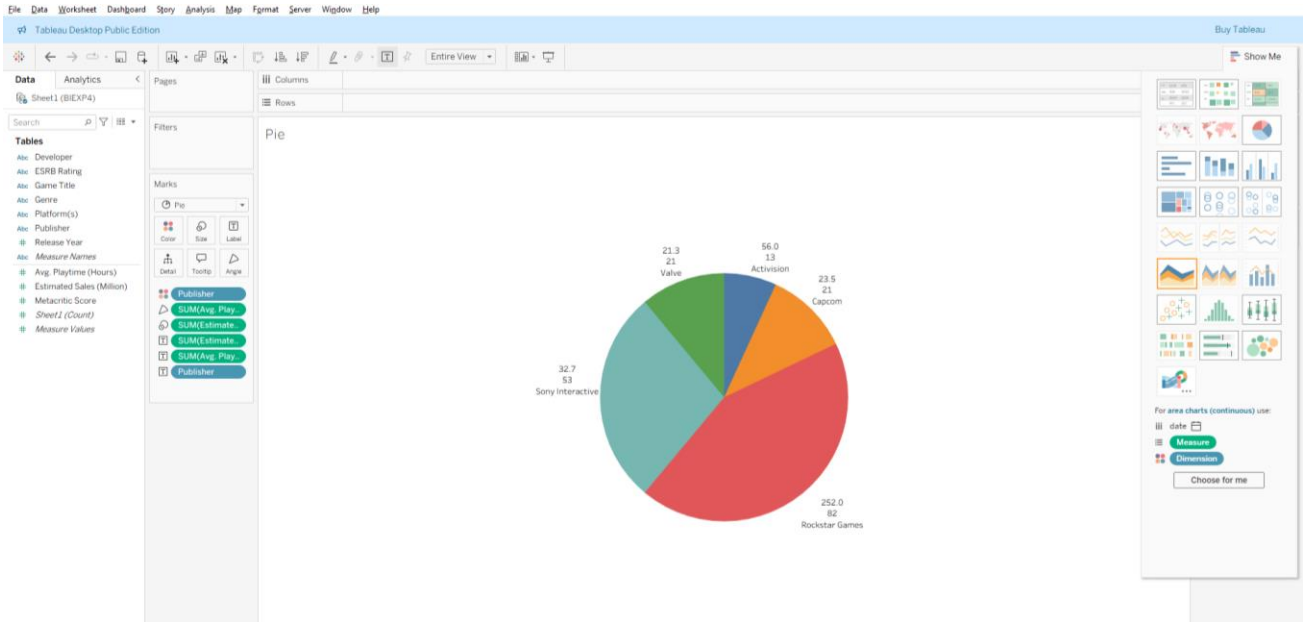
Q3. List Applications of Tableau.

- **Business Intelligence & Reporting:** Automating and distributing KPI dashboards for sales, finance, and operations.
- **Data Exploration & Discovery:** Allowing analysts to interactively explore datasets to find hidden patterns and insights.
- **Performance Monitoring:** Tracking real-time metrics across departments like marketing campaigns or website traffic.
- **Geospatial Analysis:** Creating maps to visualize regional sales, delivery routes, or demographic data.
- Trend Forecasting: Using built-in statistical models to predict future sales, inventory needs, or customer behaviour.
- **Executive Dashboards:** Providing leadership with a high-level, interactive overview of organizational health.
- **Public Data Storytelling:** Governments and NGOs use it to make complex data (e.g., budgets, census info) publicly accessible and understandable.

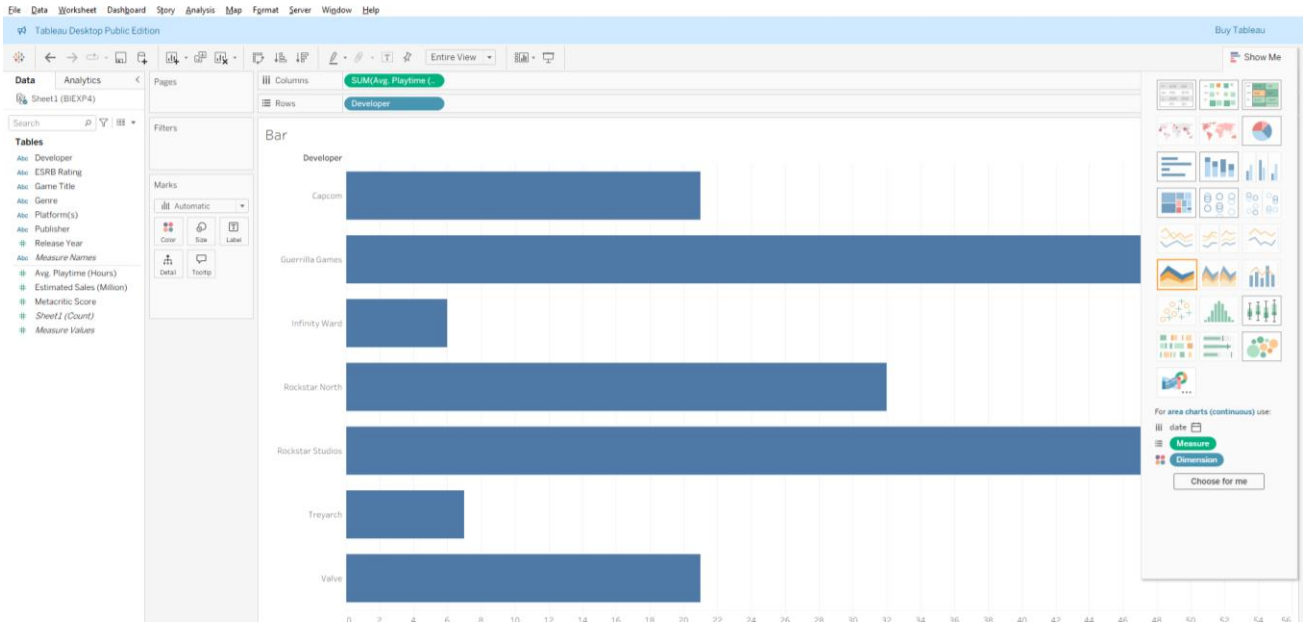
Q4. Which Data Set is being used?

Game Title	Release Year	Developer	Publisher	Genre	Platform(s)	Metacritic	Estimated	ESRB Rating	Avg. Playtime
Red Dead Redemption 2	2018	Rockstar Studios	Rockstar Games	Action-Adventure, Open	PS4, Xbox One, PC, Stadia	97	57	M	50
Grand Theft Auto V	2013	Rockstar North	Rockstar Games	Action-Adventure, Open	PS4, Xbox One, PC, PS5, XSX	96	195	M	32
Resident Evil 7: Biohazard	2017	Capcom	Capcom	Survival Horror	PS4, Xbox One, PC, Switch	86	13	M	10
Resident Evil Village	2021	Capcom	Capcom	Survival Horror, Action	PS4, PS5, Xbox, PC	84	10.5	M	11
Call of Duty: Black Ops III	2015	Treyarch	Activision	First-Person Shooter	PS4, Xbox One, PC	81	26	M	7
Call of Duty: Modern Warfare	2019	Infinity Ward	Activision	First-Person Shooter	PS4, Xbox One, PC	80	30	M	6
Half-Life	1998	Valve	Valve	FPS, Sci-Fi	PC, PS2, Linux, macOS	96	9.3	M	13
Left 4 Dead	2008	Valve	Valve	FPS, Survival Horror	PC, Xbox 360	89	12	M	8
Horizon Forbidden West	2022	Guerrilla Games	Sony Interactive	Action RPG, Open World	PS4, PS5, PC	88	8.4	T	30
Horizon Zero Dawn	2017	Guerrilla Games	Sony Interactive	Action RPG, Open World	PS4, PC	89	24.3	T	23

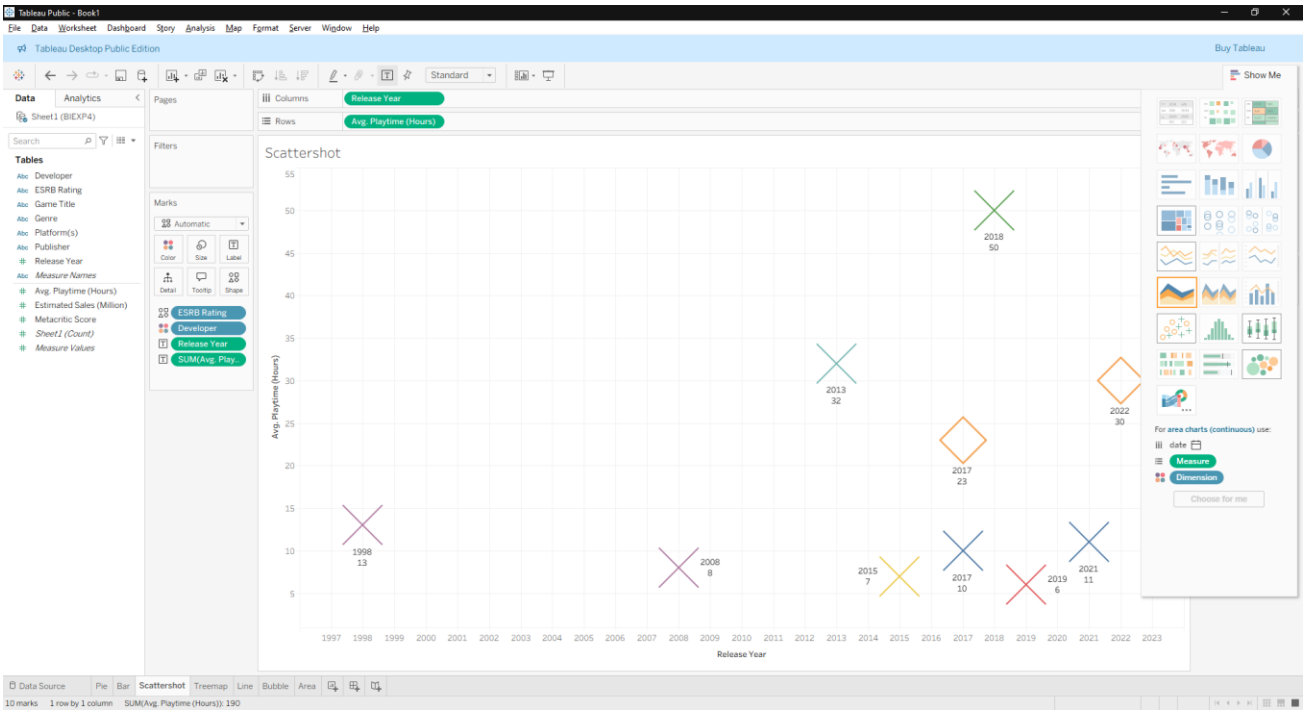
Pie Chart: (Publisher X Average Playtime X Estimated Sales)



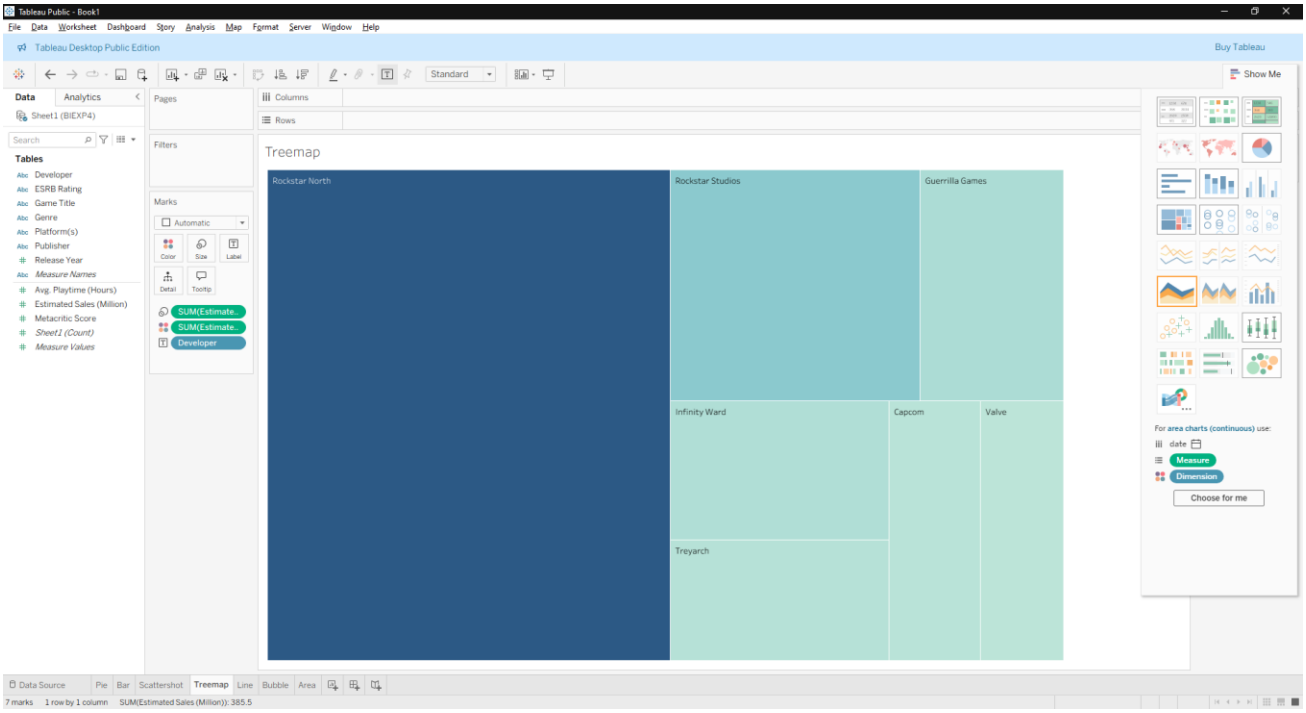
Horizontal Bar Graph: Average Playtime X Developer



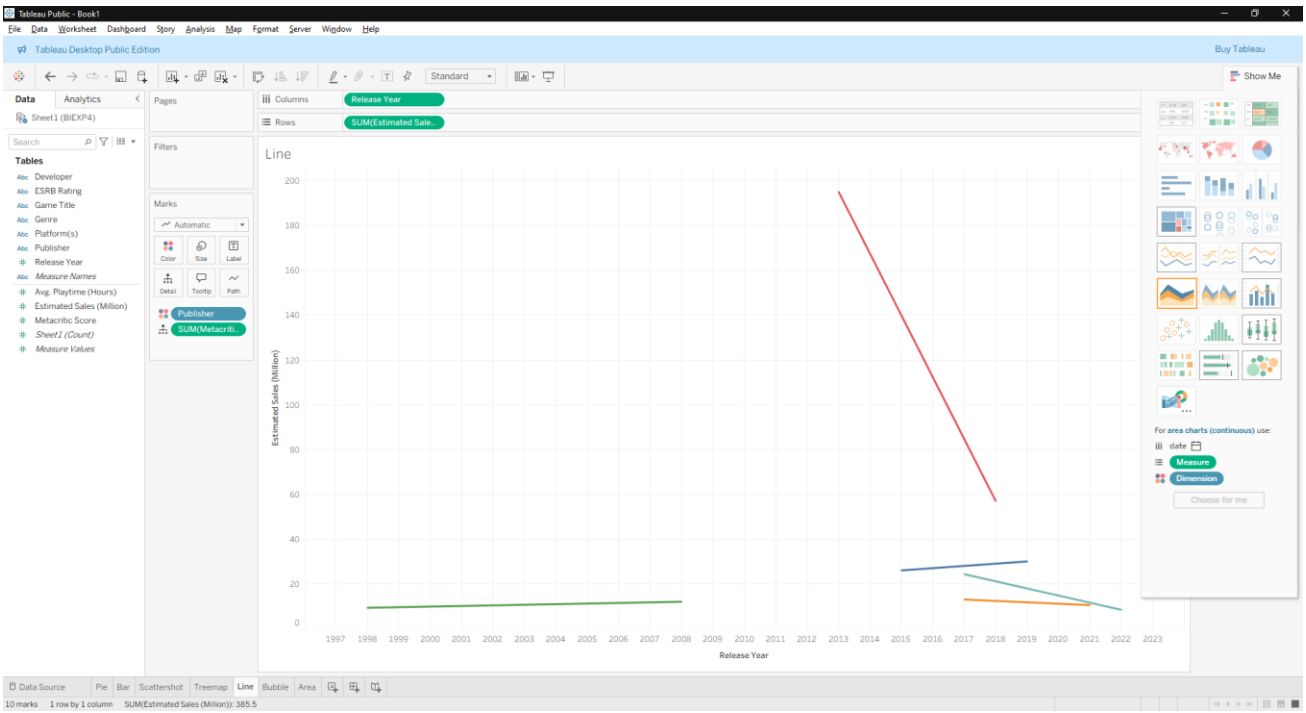
Scattershot: Release Year X Average Playtime



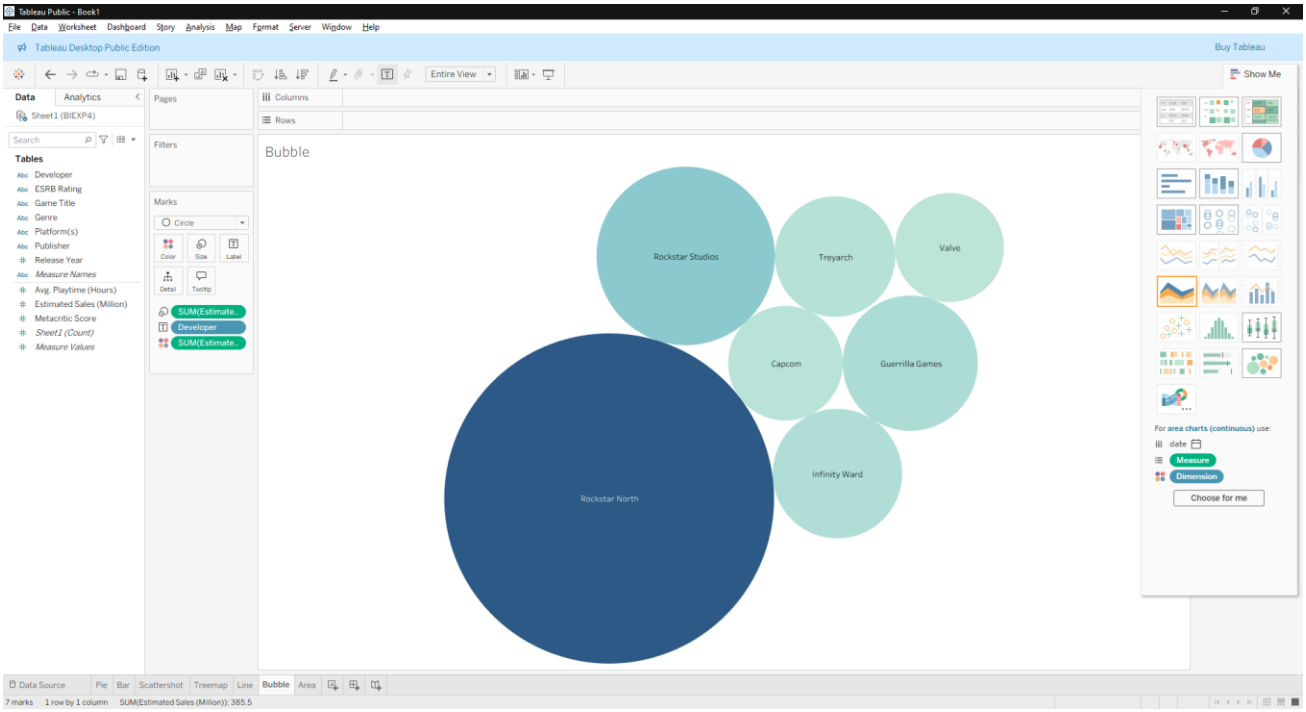
Tree map: Developer X Estimated Sales



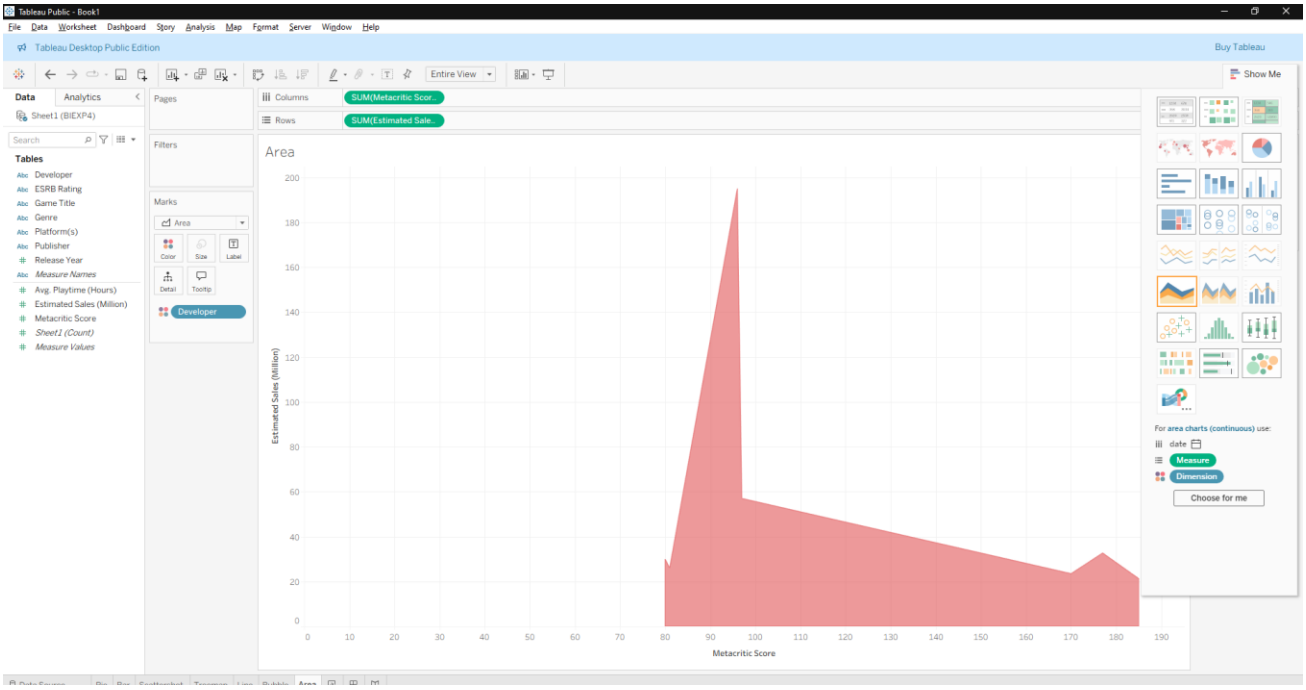
Line Chart: Release Year X Estimated Sales



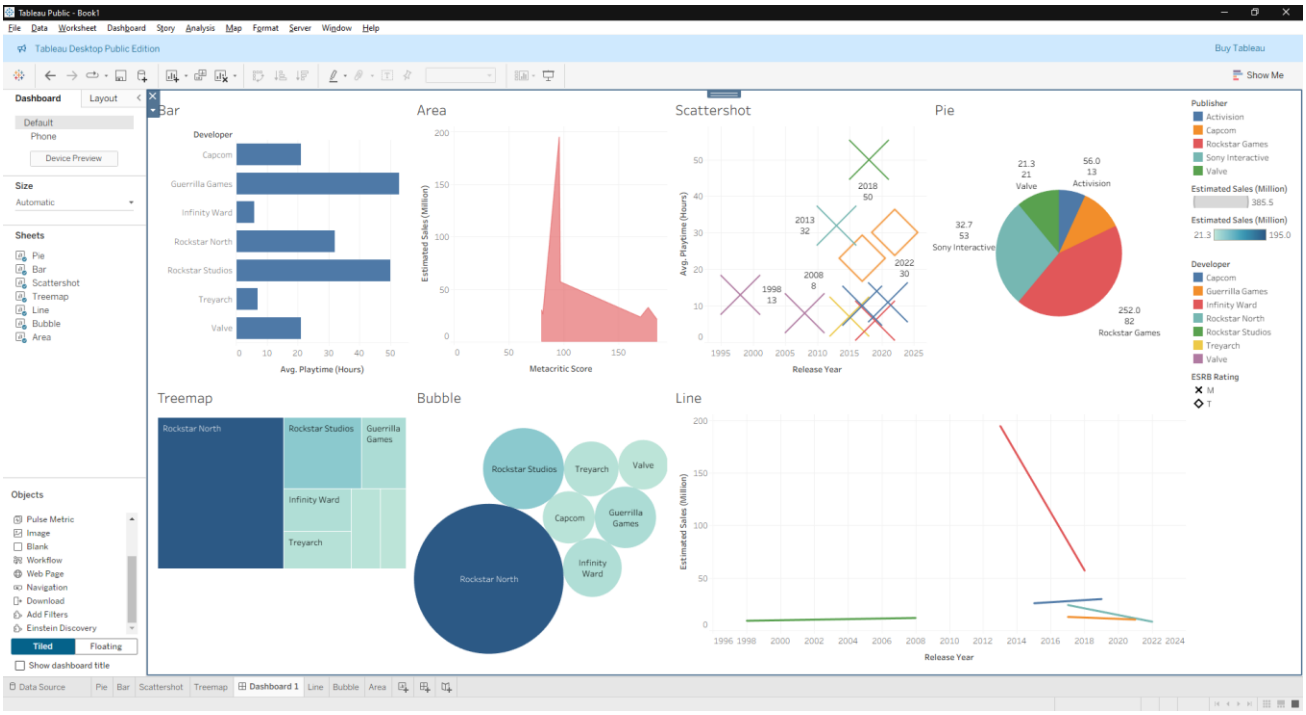
Bubble Chart: Developer X Estimated Sales



Area Continuous Graph: Metacritic Score X Estimated Sales



Custom Dashboard of all graphs used:



Conclusion

Tableau effectively transforms raw data into actionable insights through interactive visualizations and dashboards, empowering users to identify trends, make informed decisions, and communicate findings clearly and efficiently.